

Real-Time Terrain Classification for Mobile Robots Using Camera and IMU Sensor Fusion

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Abstract—Terrain awareness is an important capability for autonomous mobile robots operating in dynamic environments. Different terrains such as flat surfaces, rough paths, and slopes generate unique visual patterns and vibration characteristics that can be utilized for classification. This paper presents a real-time terrain classification system using camera and Inertial Measurement Unit (IMU) sensor fusion implemented on a Waveshare RasProver mobile rover powered by Raspberry Pi 5. Terrain data consisting of synchronized video recordings and IMU readings were collected for three terrain classes: Flat, Rough, and Slope. Videos were converted into image datasets using OpenCV while IMU values were synchronized with extracted image frames. A multimodal deep learning model combining Convolutional Neural Networks (CNN) for image feature extraction and dense neural network layers for IMU processing was developed for terrain classification. The proposed system demonstrates the feasibility of real-time terrain perception using low-cost embedded hardware and sensor fusion techniques.

Index Terms—Terrain Classification, Sensor Fusion, Mobile Robotics, IMU, Raspberry Pi 5, OpenCV, CNN, Deep Learning

I. INTRODUCTION

Autonomous mobile robots are increasingly being used in surveillance, industrial automation, planetary exploration, warehouse logistics, agriculture, disaster response, and smart mobility applications. As robotic systems continue to operate in dynamic and unstructured environments, terrain awareness becomes an important requirement for ensuring reliable mobility, stability, and navigation performance. Different terrain surfaces such as flat roads, rough paths, slopes, gravel, and uneven environments generate unique visual textures and vibration characteristics that directly influence robotic motion and control. Therefore, identifying terrain types in real time is an important challenge in robotics, autonomous systems, and intelligent mobility applications.

Traditional mobile robotic systems often rely only on predefined control strategies or visual perception techniques without dynamically considering terrain conditions during operation. Such approaches may reduce navigation efficiency and can negatively affect robot stability, wheel traction, and energy consumption when operating on varying terrains. However, terrain surfaces naturally produce both visual and inertial patterns that can be analyzed using machine learning and sensor fusion techniques for improved terrain understanding. Camera-based systems provide rich visual information regarding surface

appearance, texture, inclination, and environmental structure, while Inertial Measurement Unit (IMU) sensors capture acceleration, vibration, and motion characteristics associated with terrain interaction.

Recent advancements in embedded computing platforms such as Raspberry Pi and lightweight deep learning architectures have enabled the implementation of intelligent robotic perception systems on low-cost mobile robots. In particular, multimodal sensor fusion approaches combining camera and IMU data have shown significant potential for improving terrain classification reliability and robustness compared to single-sensor systems. This work focuses on developing a real-time terrain classification system using synchronized camera and IMU sensor fusion implemented on a Waveshare RasProver mobile rover powered by Raspberry Pi 5. The proposed system integrates synchronized dataset generation, multimodal deep learning, and real-time terrain prediction to demonstrate the feasibility of lightweight terrain-aware robotic perception using low-cost embedded hardware.

II. LITERATURE SURVEY

Lacroix et al. proposed an autonomous rover navigation system for unknown terrains using onboard sensing and robotic perception techniques. The work mainly focused on autonomous navigation and terrain understanding for planetary exploration rovers. Their study demonstrated the importance of terrain perception in improving robotic mobility and navigation efficiency in unstructured environments [1]. Angelova et al. presented a terrain analysis approach using visual information for predicting terrain slip conditions. Their work utilized camera-based terrain perception and machine learning techniques to estimate terrain interaction properties. The study highlighted the effectiveness of computer vision for identifying terrain characteristics in robotic systems [2].

Filitchkin and Byl developed a feature-based terrain classification system for the LittleDog robotic platform. The proposed approach extracted terrain features from sensor data and used classification algorithms for identifying terrain categories. Their work demonstrated that terrain classification can improve robotic locomotion and mobility adaptation [3]. Hapbold et al. proposed a real-time terrain classification and obstacle detection framework using stereo vision sensors.

Their system primarily relied on visual information for terrain perception and obstacle identification. The work demonstrated the feasibility of real-time robotic perception using vision-based approaches [4].

Brooks and Iagnemma introduced a vibration-based terrain classification system for planetary exploration rovers using accelerometer data. Their work focused on analyzing vibration signatures generated during rover movement across different surfaces. The study showed that IMU and vibration signals can effectively represent terrain properties [5]. Bengio et al. discussed representation learning techniques and their applications in machine learning and deep neural networks. Their work provided important theoretical foundations for feature extraction and multimodal learning approaches commonly used in computer vision and robotic perception systems [6]. Krizhevsky et al. introduced deep Convolutional Neural Networks (CNNs) for image classification tasks.

Their work demonstrated the effectiveness of deep learning architectures for extracting visual features from images and significantly influenced modern computer vision-based robotic perception systems [7]. Ojeda et al. proposed a terrain classification framework for mobile robots using inertial sensing and machine learning techniques. Their work utilised accelerometer-based vibration analysis to identify terrain characteristics during robot locomotion. The study demonstrated that inertial sensing can effectively distinguish terrain types while maintaining low computational complexity suitable for embedded robotic systems [8].

A. Research Gap and Novelty

Most existing terrain classification systems rely either on vision-based approaches or vibration-based approaches independently. Vision-only systems are highly dependent on lighting conditions and surface appearance, while IMU-only systems may lack sufficient contextual information for accurate terrain understanding. Additionally, several existing approaches require computationally intensive architectures or expensive robotic platforms, limiting their real-time embedded deployment capability. The proposed work addresses these limitations by developing a lightweight multimodal terrain classification framework using synchronized camera and IMU sensor fusion on a low-cost Raspberry Pi based rover platform. Unlike previous works which primarily focus on either visual or inertial sensing independently, this work combines both modalities to improve terrain perception robustness and classification reliability. Another important novelty of this work is the implementation of a complete real-time terrain prediction pipeline including synchronized dataset generation, image and IMU preprocessing, multimodal deep learning, and embedded deployment capability. The proposed system demonstrates that real-time terrain classification can be achieved using low-cost hardware and lightweight deep learning architectures suitable for mobile robotic systems.

III. OBJECTIVE

The primary objective of this work is to develop a real-time terrain classification system for autonomous mobile robots using camera and IMU sensor fusion. The proposed system aims to improve robotic terrain perception by combining visual information obtained from onboard cameras with inertial measurements obtained from an Inertial Measurement Unit (IMU). The project focuses on identifying different terrain categories such as Flat, Rough, and Slope terrains using synchronized sensor inputs and deep learning techniques. Another important objective of this work is to create a synchronized multimodal dataset consisting of terrain images and corresponding IMU readings. The collected data is processed and organized into structured datasets which can be used for training machine learning models. The work also aims to implement real-time prediction capability on an embedded robotic platform using Raspberry Pi 5 and lightweight deep learning architectures suitable for mobile robotics applications. The proposed system further demonstrates the feasibility of low-cost robotic perception systems using sensor fusion techniques. The developed framework can later be extended for intelligent navigation, terrain-aware control systems, and autonomous robotic mobility in dynamic environments.

IV. METHODOLOGY

The proposed terrain classification system consists of multiple stages including hardware setup, data collection, dataset preparation, sensor synchronization, model training, and real-time prediction. The complete system was implemented using a Waveshare RasProver mobile rover integrated with Raspberry Pi 5, onboard camera module, and IMU sensor.

A. Hardware Setup



Fig. 1. RasProver used for terrain data collection and real-time prediction

The hardware platform used for this work is the Waveshare RasProver mobile rover powered by Raspberry Pi 5. The rover consists of a four-wheel differential drive chassis integrated with onboard sensing and processing hardware. A camera module mounted on the rover is used for capturing terrain visuals during movement, while the onboard IMU sensor records acceleration and vibration characteristics generated by different terrain surfaces. The Raspberry Pi 5 acts as the main embedded processing unit responsible for sensor interfacing, dataset generation, model inference, and real-time terrain prediction. The rover is also integrated with a wireless controller interface which enables manual navigation during terrain data collection. The combination of camera and IMU sensors allows the system to capture both visual and inertial information associated with terrain characteristics.

B. Data Collection

Terrain data collection was performed for three terrain categories: Flat terrain, Rough terrain, and Slope terrain. For each terrain class, multiple video recordings were captured using the onboard camera while the rover was moving across the corresponding surface. Simultaneously, IMU sensor readings were continuously recorded at approximately 30 Hz to capture terrain-induced vibrations and acceleration patterns. The IMU dataset included acceleration measurements along the X, Y, and Z axes along with calculated vibration magnitude values. Two video recordings were collected for each terrain class and later converted into image datasets for model training. Care was taken to ensure proper synchronization between camera frames and IMU readings during data acquisition.

C. Dataset Preparation

The recorded terrain videos were converted into image frames using OpenCV-based preprocessing techniques. Frames were extracted periodically from each video to generate image datasets corresponding to the three terrain categories. Approximately 800 images were generated for each terrain class. The IMU readings were synchronized with extracted image frames using timestamp mapping techniques. For each saved image frame, the nearest IMU reading was selected and mapped to the corresponding image. The synchronized data was then organized into structured image and IMU datasets which were later used for multimodal model training. The final dataset structure consisted of separate folders for Flat, Rough, and Slope terrain classes containing synchronized image data and IMU CSV files. This organization simplified model training and preprocessing operations.

D. Dataset Split Strategy

To improve model generalization capability and reduce overfitting, the collected terrain dataset was divided into separate training and testing datasets. Multiple terrain recordings captured under different environmental conditions were used for dataset generation. Frames extracted from different videos were randomly distributed into train and test datasets to

ensure variation in terrain appearance, lighting, and vibration characteristics. Dataset balancing techniques were also applied during training to prevent class imbalance and improve classification fairness across Flat, Rough, and Slope terrain categories.

E. Sensor Fusion Model

The terrain classification model was developed using a multimodal deep learning architecture combining image-based and inertial-based feature extraction. A Convolutional Neural Network (CNN) architecture was used to extract terrain features from camera images while dense neural network layers processed IMU sensor values. The image branch of the network learns visual terrain patterns such as texture, inclination, and surface appearance, whereas the IMU branch learns vibration and acceleration characteristics associated with terrain motion. Both feature vectors are combined using sensor fusion layers and passed through fully connected layers for final terrain classification. The model was implemented using TensorFlow and OpenCV libraries. The final output layer classifies the terrain into one of the three categories: Flat, Rough, or Slope.

F. Training Configuration

The multimodal terrain classification model was trained using TensorFlow on synchronized image and IMU datasets. MobileNetV2 pretrained on ImageNet was used as the backbone feature extractor for image processing. The model was trained using the Adam optimizer with categorical cross-entropy loss. Data augmentation techniques including random flipping, rotation, and zooming were applied to improve model robustness. Early stopping, learning rate reduction, and checkpoint saving techniques were used to improve training stability and prevent overfitting. Dataset balancing techniques were also implemented to ensure fair training across all terrain classes.

G. Real-Time Prediction

During real-time operation, live camera frames and continuous IMU sensor readings are acquired from the rover. The camera images are preprocessed and resized before being passed into the CNN branch of the model. Simultaneously, IMU values are processed through the inertial feature extraction branch. The trained fusion model combines both inputs and predicts the terrain type dynamically in real time. The predicted terrain class is displayed continuously during rover operation. This real-time prediction pipeline demonstrates the feasibility of sensor fusion-based terrain perception on embedded robotic platforms using lightweight deep learning models.

H. Simulation Environment

A simulation environment was also developed using the PyBullet physics engine to demonstrate terrain-aware robotic navigation based on terrain classification results. The simulated environment consisted of multiple terrain regions representing Flat, Rough, and Slope surfaces. Different terrain types were represented using distinct surface colors where

white represented smooth flat terrain, brown represented rough terrain, and green represented slope terrain.

The mobile robot was programmed to navigate autonomously from a start point to a target point using path planning techniques. During navigation, the terrain classification system continuously identified the terrain type and dynamically adjusted the rover velocity according to the detected terrain condition. Higher speeds were maintained on smooth flat terrains while reduced speeds were applied on rough and slope terrains to improve navigation stability and safety.

The simulation demonstrated the practical applicability of multimodal terrain classification for intelligent robotic mobility and adaptive navigation systems. The PyBullet-based simulation environment also validated the integration of terrain perception with terrain-aware motion control strategies for autonomous mobile robots.

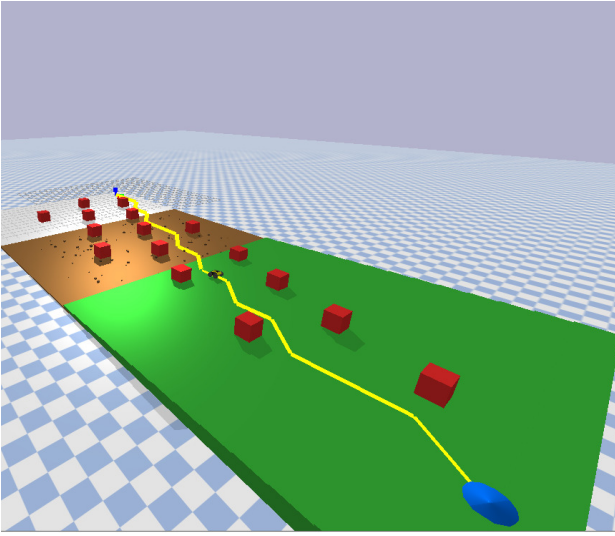


Fig. 2. PyBullet simulation environment demonstrating terrain-aware navigation and adaptive speed control

V. RESULTS AND DISCUSSION

The proposed terrain classification system successfully generated synchronized multimodal datasets consisting of terrain images and corresponding IMU sensor readings. The data collection and preprocessing pipeline enabled proper synchronization between visual and inertial information, allowing the creation of structured datasets suitable for multimodal machine learning applications. The developed fusion model was capable of learning terrain-specific visual patterns and vibration characteristics from the synchronized dataset. The synchronized dataset generation process also ensured proper temporal alignment between visual frames and inertial sensor measurements, improving the consistency and reliability of multimodal feature learning.

Multiple terrain recordings were collected for Flat, Rough, and Slope surfaces using the onboard rover camera and IMU sensor. Videos were converted into image frames using OpenCV and synchronized with corresponding IMU readings

to create a multimodal dataset. The dataset was organized into separate training and testing folders to evaluate model generalization performance. Dataset balancing and random train-test distribution techniques were also implemented to reduce class imbalance and improve prediction reliability. Different terrain environments and surface appearances were included during data collection to improve model robustness against environmental variations and reduce overfitting to specific terrain textures.

The terrain classification model was trained using synchronized camera images and IMU sensor readings collected from Flat, Rough, and Slope terrains. MobileNetV2 based transfer learning combined with image-dominant sensor fusion significantly improved the classification performance. Data augmentation, IMU normalization, dataset balancing, and frame redundancy reduction techniques further improved model generalization capability and reduced overfitting. Transfer learning enabled efficient extraction of high-level visual features from terrain images, while the IMU branch contributed additional vibration and acceleration-based terrain characteristics. The fusion of both modalities improved the robustness of terrain understanding compared to single-sensor approaches. The developed architecture also demonstrated efficient computational performance suitable for deployment on embedded robotic hardware platforms such as Raspberry Pi.

The proposed model achieved high terrain classification accuracy on the test dataset. The classification report demonstrated strong precision, recall, and F1-score values for all terrain classes, indicating reliable multimodal terrain prediction performance. The overall classification accuracy achieved by the proposed system was approximately 95.67%. The Flat terrain class achieved high recall due to consistent surface characteristics, while the Rough terrain class achieved perfect classification performance during testing. The Slope terrain class also demonstrated strong classification capability despite variations in terrain inclination, rover vibration patterns, and camera perspective. These results indicate that the proposed multimodal approach was capable of effectively learning both visual and inertial terrain characteristics for accurate classification.

TABLE I
MODEL PERFORMANCE METRICS

Class	Precision	Recall	F1-Score	Support
Flat	0.92	1.00	0.96	134
Rough	1.00	1.00	1.00	39
Slope	1.00	0.86	0.93	81

The confusion matrix generated during testing demonstrated that the trained model was able to correctly distinguish terrain categories with very low misclassification rates. The integration of IMU sensor data improved terrain understanding by providing vibration and acceleration information in addition to visual terrain appearance. The generated heatmap further validated the effectiveness of the multimodal fusion approach for terrain classification tasks. The low number of misclassified

samples indicates that the model successfully learned discriminative terrain features from the synchronized multimodal dataset.

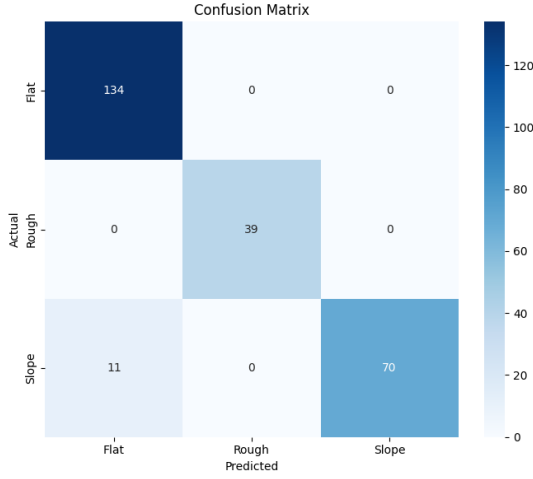


Fig. 3. Confusion Matrix Heatmap of Terrain Classification Model

The real-time prediction pipeline was also successfully implemented using live camera feed and continuous IMU readings from the rover platform. The system demonstrated the capability to perform terrain prediction dynamically during rover movement using low-cost embedded hardware. Additionally, a PyBullet simulation environment was developed to demonstrate terrain-aware robotic navigation and adaptive speed control based on terrain classification outputs. In the simulation environment, the robot autonomously navigated between target locations while dynamically adjusting movement speed according to detected terrain conditions. The achieved experimental and simulation results validate the feasibility of lightweight multimodal terrain classification for intelligent mobile robotic systems using Raspberry Pi based embedded deployment.

VII. FUTURE SCOPE

The proposed terrain classification framework can be further improved by increasing the dataset size and introducing additional terrain categories such as grass, sand, gravel, and uneven outdoor surfaces. A larger and more diverse dataset would improve model generalization capability and robustness under varying environmental conditions. Future work can also focus on optimizing the deep learning architecture for faster inference on embedded hardware platforms. Lightweight neural network architectures and model compression techniques can be implemented to improve computational efficiency and reduce latency during real-time prediction. Another important extension of this work is the integration of terrain-aware autonomous navigation systems. The terrain classification output can be combined with robotic control algorithms for intelligent path planning and adaptive mobility systems. The developed framework can also be integrated with SLAM systems, obstacle avoidance algorithms, and autonomous exploration robots

for advanced robotic perception and navigation applications. Further research can also investigate advanced sensor fusion architectures, reinforcement learning-based terrain adaptation, and multi-sensor robotic perception systems for autonomous field robotics and industrial automation applications.

VIII. CONCLUSION

This work presents a real-time terrain classification system using camera and IMU sensor fusion implemented on a Raspberry Pi based mobile rover platform. The proposed framework combines visual terrain information from onboard cameras with inertial measurements obtained using an IMU sensor to perform multimodal terrain perception. A complete pipeline was developed including synchronized terrain data collection, image extraction, IMU synchronization, dataset preparation, multimodal deep learning model training, and real-time terrain prediction.

The terrain classification model was developed using MobileNetV2 based transfer learning for image feature extraction along with dense neural network layers for IMU feature processing. Sensor fusion techniques were used to combine visual and inertial information for improved terrain classification reliability. Experimental results demonstrated high classification performance for Flat, Rough, and Slope terrains, validating the effectiveness of the proposed multimodal approach.

The developed system also demonstrated the feasibility of performing real-time terrain prediction on low-cost embedded robotic hardware using lightweight deep learning architectures. The proposed framework provides a scalable, computationally efficient, and cost-effective solution for intelligent robotic perception and future terrain-aware autonomous navigation systems.

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